Spiking Data Preprocessing Guide

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Code File Appendix

This is an appendix with the important coding files in the spike processing pipeline. There are more functions and files used, but these are the most important and the ones that you will likely make changes to based on your use case.

Neural-Data-Reading/Processing_Functions_cgg

All Data Processing

- DATA_cggAllSessionInformationConfiguration.m
- DATA_pgpAllSessionSpikeProcessing_v2.m

Parameters

- PARAMETERS_cgg_getSessionProbeInformation.m
- PARAMETERS_cgg_procFullTrialPreparation_v2.m

Neural_Data_cgg

- pgp_proc_EventFrameTrialPreparation.m
- pgp_procProcessingBehaviorOnly.m
- pgp_buildAlignFromEpoch.m

Input_Output_Information

- pgp_exportCfgToJson.m

Spike_Alignment

- pgp_spikeTrialAlignment_v4.py

Step 1 - Prerequisites

1. In DATA_cggAllSessionInformationConfiguration.m and PARAMETERS_cgg_getSessionProbeInformation.m, add in the session names and the probe channels and numbers for your respective experiment and monkey.
2. In PARAMETERS_cgg_procFullTrialPreparation_v2.m choose your alignment event and baseline parameters. Currently, this should be set to:

```
case 'Decision' Frame_Event_Selection_Data = 'SelectObject';
Frame_Event_Selection_Location_Data = 'END';
Frame_Event_Selection_Baseline = 'Blink';
Frame_Event_Selection_Location_Baseline = 'START'; end
Window_Before_Data = 1.5; Window_After_Data = 1.5; % 500 ms
after blink in start of trial Window_Before_Baseline = 0;
Window_After_Baseline = 0.5; % might also be something like
below: % less likely to go into object selection during trial %
Window_Before_Baseline = 0.15; % Window_After_Baseline = 0.4;
```

This will segment trials around when the object was recorded as selected in Unity which means that the monkey fixation started 700ms before this. Object selection will be at the middle of the trial aka timepoint 1500/3000 given 1.5 seconds before and after.

3. Make sure to have a python3 virtual environment with the necessary python packages to run the pgp_spikeTrialAlignment_v4.py script installed (numpy, pandas, scipy, check file for any other packages).
4. You will also need the sorted KS4 data from David and the <session_name>_BHV folders and potentially the record node folders from DATA_Neural (original recording files)

Step 2 - Main Code

The main code runs the master file: DATA_pgpAllSessionSpikeProcessing_v2.m

In this, 5 main scripts are executed

1. pgp_proc_EventFrameTrialPreparation.m which creates trialDef and gameframedata files with the recording times for each trial and event
2. pgp_precprocessingBehaviorOnly.m which segments trials and gets your trial variables file from pgp_getTrialVariables. The trial variables file contains all the info about the Trial number, trial in block number, block, object selected, selected location, etc... To modify this change the pgp_getTrialVariables script.
3. pgp_buildAlignFromEpoch.m creates an alignment file that that is used to align the spike times to each trial. This is run for both the task data and the baseline data.
4. pgp_exportCfgToJson.m exports information about the session from DATA_cggAllSessionInformationConfiguration.m and info about the segmentation parameters and other details from PARAMETERS_cgg_procFullTrialPreparation_v2.m. You will have to point the script to where you want this saved.
5. pgp_spikeTrialAlignment_v4.py is run through the matlab terminal which lines up the spike times with the Salign file generated and segments all trials and baselines and generates your spiking files. It also does some quality assurance checks to make sure that all the spike times in your aligned files are correct.

Things You Need to do to run:

1. Have your python environment running in the terminal.
2. point json_out to where you want the json saved to
3. point sorted_root_base to where the sorted spike files are. Currently they are both on TEBA and on ACCRE under a folder called 'M1VU595_ACC_PFC_CD/sorted'

4. make sure that the arguments in the system call which runs the spike trial alignment match the parameters used for trial segmentation. e.g. make sure that —win_before, and —win_after reflect the parameters used to get trial time during the task and —win_before_base and —win_after_base have the correct before and after times from PARAMETERS_cgg_procFullTrialPreparation_v2.m
5. Run the script

Part 3 Code Output

The output of this will be a .mat in a folder called SpikeAligned called <session_name>_SpikeCounts.mat

In this is a cell array called X that has an entry for each trial with a cell that has a matrix that is num_neurons x num_timepoints as well as X_base which has the corresponding baseline. TrialId is simply the number of cells in X aka the number of trials. It should correspond precisely to the number of rows in the TargetInformation.mat file which can be found in Session/Epoched_Data/Decision/Target. Meta contains info such as the sampling frequency, how much time there was before and after event and baseline, number of probes, and info about each unit such as what probe it belongs to which is important for sorting units into their respective areas.